

NIFTY 100

FINANCIAL INTELLIGENCE PLATFORM

Analyst Guide

Dashboard • Analytics • Reports • REST API • Testing

This guide explains how an analyst can use the NIFTY 100 Financial Intelligence Platform to explore company financials, screen companies, compare peers, review sectors, inspect trends, evaluate capital allocation and access generated reports.

The platform combines a SQLite analytical database, Python financial analytics, a Streamlit dashboard, a FastAPI service and automated tests.

1. Platform Overview

The platform is organized into several layers. Raw financial datasets are processed through the ETL layer and stored in data/nifty100.db. Analytical modules calculate financial ratios, cash-flow metrics, valuation outputs, portfolio statistics and clustering results.

Layer	Purpose
Data / ETL	Loads, normalizes and validates financial datasets.
Database	Stores company, financial, market, peer and document data.
Analytics	Calculates KPIs, ratios, CAGR, valuation and cash-flow metrics.
Clustering	Groups companies into five financial archetypes.
API	Exposes analytical data through FastAPI endpoints.
Dashboard	Provides interactive Streamlit screens for analysts.
Reports	Provides company tearsheets, sector reports and portfolio reports.
Testing	Validates ETL, analytics, APIs, dashboard and performance.

Main database

data/nifty100.db

Main application commands

```
python -m streamlit run src/dashboard/app.py
uvicorn src.api.main:app --reload --port 8000
pytest -q
```

2. Dashboard Navigation

The Streamlit application contains eight screens. They are implemented under `src/dashboard/pages` and are accessed from the Streamlit navigation.

Screen	File	Primary purpose
Home	01_home.py	Platform overview and key information.
Profile	02_profile.py	Company-level financial profile.
Screener	03_screener.py	Filter and inspect companies.
Peers	04_peers.py	Compare companies with peer groups.
Trends	05_trends.py	Inspect financial trends over time.
Sectors	06_sectors.py	Analyze sector-level information.
Capital	07_capital.py	Review capital allocation and cash-flow information.
Reports	08_reports.py	Access generated reports and documents.

Starting the dashboard

```
python -m streamlit run src/dashboard/app.py
```

After starting Streamlit, use the navigation controls to move between screens. The dashboard reads analytical information from the SQLite database and generated report files.

3. Home and Company Profile

Home screen

The Home screen is the starting point for the platform. It provides a high-level view of the analytical application and its available modules.

- Use Home when first opening the dashboard.
- Use the navigation menu to move into company and analytical screens.
- Use the reports and analytics screens for deeper investigation.

Company Profile screen

The Profile screen is used for company-level investigation. An analyst can select a company and inspect information sourced from the company master data and related financial tables.

- Company identity and descriptive information
- Sector and industry classification
- Financial history
- Financial ratios
- Valuation information
- Peer information
- Available company documents

Analyst workflow

A typical workflow is to begin with a company profile, inspect historical financials and ratios, then move to peers, trends and generated reports for additional context.

4. Screener

The Screener screen provides a structured way to narrow the company universe using financial criteria. The corresponding API endpoint is GET /api/v1/screener/.

Available screening dimensions

- Sector
- Minimum return on equity
- Maximum debt-to-equity
- Minimum five-year revenue CAGR
- Minimum operating profit margin
- Minimum free cash flow

Using the screener

Select the desired filters, review the returned company set and then open individual companies in the Profile screen for detailed inspection.

API equivalent

```
curl "http://127.0.0.1:8000/api/v1/screener/?min_roe=15"
```

The screener should be treated as a filtering and exploration tool. Screening results reflect the data and criteria supplied to the platform and should be investigated further through the underlying financial data.

5. Peer Comparison and Trends

Peer Comparison

The Peer Comparison screen uses the platform's peer-group information and percentile data. It is designed to provide relative context for company-level metrics.

- Select a company.
- Review its peer group.
- Inspect available peer metrics and percentile information.
- Use the company Profile screen for additional financial detail.

Peer API

```
curl http://127.0.0.1:8000/api/v1/peers/BAJAJAUTO
```

Trends

The Trends screen is used to inspect financial changes across years. Historical information can help identify changes in profitability, growth, cash flow and other financial measures.

- Compare historical periods rather than relying on a single year.
- Look for changes in direction and consistency.
- Use the underlying company financials to validate unusual movements.

6. Sector Analysis and Capital Allocation

Sector Analysis

The Sector Analysis screen organizes companies by sector and provides sector-level analytical context.

- Review the available sectors.
- Select a sector for company-level inspection.
- Use sector information to provide context around company metrics.
- Cross-check sector results with individual company data.

Sector API

```
curl http://127.0.0.1:8000/api/v1/sectors/  
curl http://127.0.0.1:8000/api/v1/sectors/Financials
```

Capital Allocation

The Capital Allocation screen focuses on cash-flow-related analytical information. The platform calculates metrics such as free cash flow, CFO/PAT ratio, CapEx intensity and related classifications.

- Review operating cash generation.
- Review free cash flow.
- Inspect capital expenditure intensity.
- Consider the relationship between accounting profit and cash generation.

7. Annual Reports and Generated Reports

Annual Reports screen

The Annual Reports screen provides access to available company documents stored in the documents dataset and generated reporting outputs.

- Search or select the relevant company.
- Review available document records.
- Use generated company tearsheets for summarized financial analysis.

Company tearsheets

Company tearsheets are stored in reports/tearsheets. The repository contains generated tearsheets for the analytical company universe.

`reports/tearsheets/_tearsheet.pdf`

Sector reports

`reports/sector/_report.pdf`

Portfolio report

`reports/portfolio/portfolio_summary.pdf`

Reports provide a convenient presentation layer over the underlying analytical outputs. For detailed investigation, analysts should also inspect the database-backed dashboard and API responses.

8. KMeans Clustering and Analytical Reports

The clustering workflow groups companies using five financial features: return on equity, debt-to-equity, five-year revenue CAGR, five-year FCF CAGR and operating profit margin.

Clustering workflow

- Prepare the latest company-level analytical features.
- Apply sector-median imputation where required.
- Standardize the features.
- Run KMeans with five clusters.
- Calculate distance from the cluster centroid.
- Profile and label the resulting clusters.

Outputs

```
output/cluster_labels.csv
output/cluster_profiles.csv
reports/elbow_plot.png
reports/correlation_heatmap.png
output/outlier_report.csv
output/portfolio_stats.csv
```

Elbow plot

The elbow plot records KMeans inertia across candidate cluster counts and is used as a diagnostic for the clustering workflow.

Correlation heatmap

The correlation heatmap provides Pearson correlation information for selected financial KPIs.

9. FastAPI Analyst Guide

The FastAPI service provides programmatic access to the platform's analytical data. Start the API from the project root.

```
uvicorn src.api.main:app --reload --port 8000
```

Interactive API documentation

```
http://127.0.0.1:8000/docs
```

Company examples

```
curl http://127.0.0.1:8000/api/v1/companies/  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/financials  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/ratios  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/valuation  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/peers  
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/documents
```

Portfolio and documents

```
curl http://127.0.0.1:8000/api/v1/portfolio/stats  
curl http://127.0.0.1:8000/api/v1/documents/
```

10. API Health, OpenAPI and Postman

Health checks

```
curl http://127.0.0.1:8000/api/v1/health/  
curl http://127.0.0.1:8000/api/v1/system/health
```

The system health endpoint provides API and database status information including table-level row counts and uptime information.

OpenAPI

The generated OpenAPI specification is stored at `reports/openapi.json`. The same specification is available interactively from the running FastAPI application's documentation.

```
reports/openapi.json
```

Postman

A Postman collection containing the API requests is available for manual endpoint testing.

```
reports/postman_collection.json
```

Recommended API validation flow

- Start the FastAPI server.
- Open `/docs`.
- Check the health endpoint.
- Test the company endpoint.
- Test the screener.
- Test portfolio statistics.
- Use Postman for repeated endpoint checks.

11. Testing and Troubleshooting

Full test suite

```
pytest -q
```

Targeted tests

```
pytest tests/test_api.py -q
pytest tests/test_integration.py -q
pytest tests/test_dashboard.py -q
pytest tests/test_performance.py -q -s
```

Common troubleshooting

- If the dashboard does not start, use 'python -m streamlit run src/dashboard/app.py'.
- If FastAPI imports fail, activate .venv and start Uvicorn from the project root.
- If database errors occur, verify that data/nifty100.db exists.
- If a test fails, run that test file separately before running the full suite.
- If an API endpoint returns unexpected data, inspect /docs and the corresponding router.

Performance validation

```
pytest tests/test_performance.py -q -s
python src/performance/check_query_plans.py
```

The performance suite covers repeated API calls, concurrent API calls and representative SQLite queries. SQLite indexes are used for commonly queried company, year, peer and document fields.

12. Analyst Workflow Checklist

The following workflow provides a repeatable approach for exploring a company or a group of companies through the platform.

Step	Action
1	Start the Streamlit dashboard.
2	Open the Home screen and review the available modules.
3	Select a company in Profile.
4	Review financial history and ratios.
5	Inspect valuation information.
6	Review peer information.
7	Use Trends to inspect historical changes.
8	Use the Screener for structured filtering.
9	Use Sector Analysis for sector context.
10	Review Capital Allocation and cash-flow information.
11	Open Annual Reports and company tearsheets.
12	Use FastAPI when programmatic access is required.
13	Validate results with the underlying database-backed outputs.

Analytical caution

Dashboard filters, rankings, clusters and financial metrics are derived from the platform's datasets and analytical rules. Results should be interpreted together with their underlying financial context rather than as standalone conclusions.

13. Project Reference Map

The following locations are useful when maintaining or extending the platform.

Location	Purpose
src/etl/	Data loading, normalization and validation.
src/analytics/	Financial KPI and analytical calculations.
src/api/	FastAPI application and routers.
src/dashboard/	Streamlit application and pages.
src/performance/	Database indexes and query-plan checks.
tests/	Automated regression and integration tests.
data/nifty100.db	Main analytical SQLite database.
output/	Machine-readable analytical outputs.
reports/	Visualizations, API specifications and generated reports.
docs/	Project documentation and analyst guide.

Useful commands

```
python -m streamlit run src/dashboard/app.py
uvicorn src.api.main:app --reload --port 8000
pytest -q
git status
```

This guide is intended as a practical reference for analysts and developers working with the NIFTY 100 Financial Intelligence Platform.